LOOPING STATEMENTS

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- We use loops to perform repeated actions
- Eg: If you are designed a task of printing numbers from 1 to 100 It will be very hectic to do it manually, loops helps us automate such tasks

- **for** - loops through a block of code a number of times
- **while** - loops through a block of code while a specified condition is true
- **do/while** - also loops through a block of code while a specified condition is true

Types of Loops in JavaScript

1. for loop – Loop a block of code number of times
2. for in loop – loops through the key of an object
3. for of loop – loops through the value of an object
4. while loop – loops a block based on specific condition
5. do-while loop –while loop variant which runs atleast once

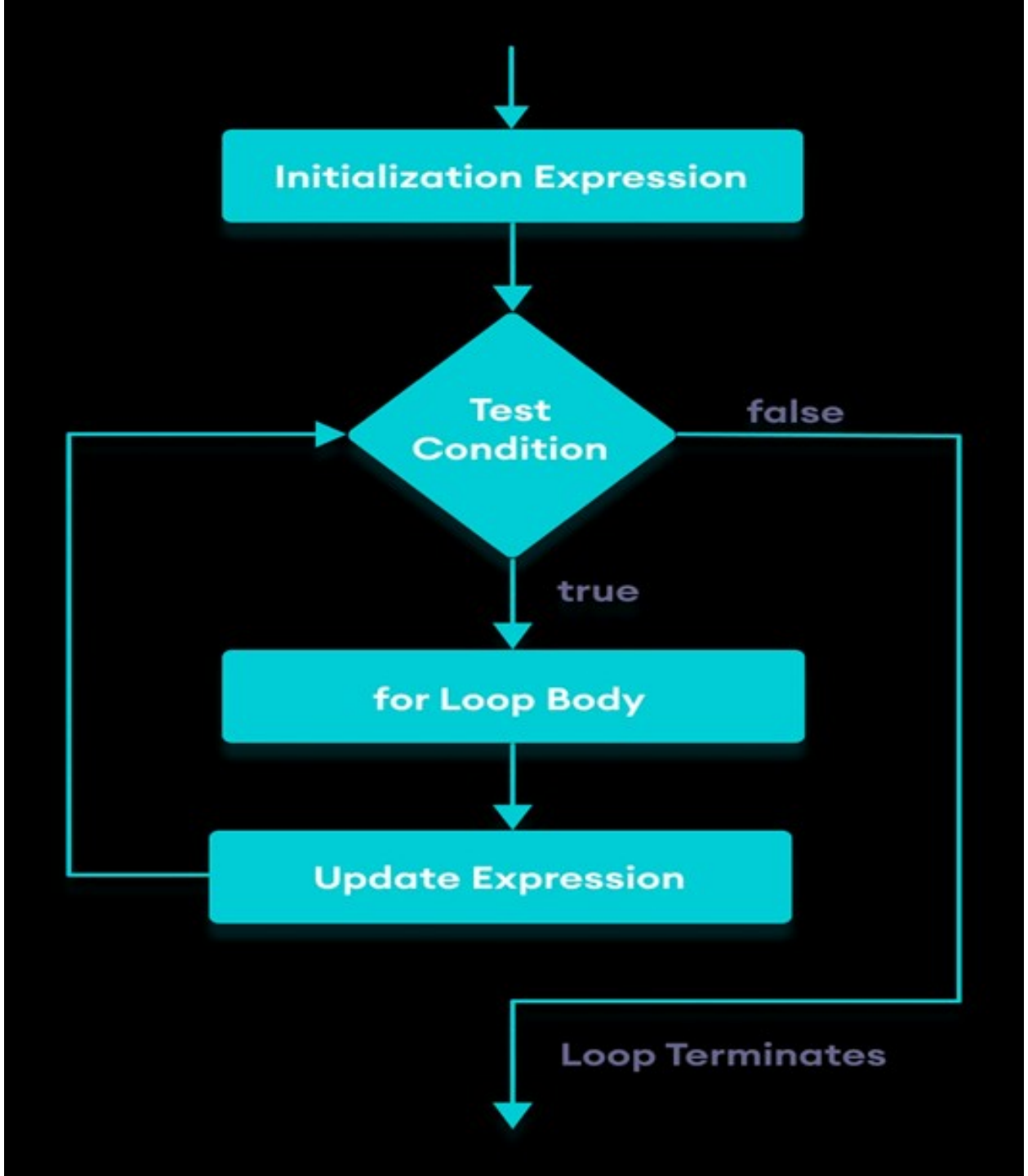
The for loop

- The syntax of the for loop is:

```
for (initialExpression; condition; updateExpression) {
   // for loop body
}
```

```
var n=5;
for (i=0;i<=n;i++){
console.log(i);
}
```

* The initialExpression initializes and/or declares variables and executes only once.
* The condition is evaluated.
* If the condition is false, the for loop is terminated.
* If the condition is true, the block of code inside of the for loop is executed.
* The updateExpression updates the value of initialExpression when the condition is true.
* The condition is evaluated again. This process continues until the condition is false.


Initialization Expression
Test Condition
false
true
for Loop Body
Update Expression
Loop Terminates

Tasks

- Example 1: Display a text in 20 times
- Example 2: Display Numbers from 1 to 5
- Example 3: Display Sum of n Natural Numbers 1 to 100

The for in loop

Syntax:

```
for (key in object) {

    // body of for...in

}
```

- Eg:

```
const student = {
    name: 'Monica',
    class: 7,
    age: 12
}

// using for...in
for ( let key in student ) {

    // display the properties
    console.log(`${key} => ${student[key]}`);
}
```

Task

- Use for...in loop is used to iterate over the student object and print all its properties.

  The object key is assigned to the variable key.

  student[key] is used to access the value of key.

Input:

```
const student={
name:’Monica’,
class:7,
age:12}
```

Output:

```
name => Monica
class => 7
age => 12
```

The for of loop

- The syntax of the for...of loop is:

```
for (element of iterable) {
    // body of for...of
}
```

- **iterable** - an iterable object (array, set, strings, etc).
- **element** - items in the iterable

```
// array
const students = ['John', 'Sara', 'Jack'];

// using for...of
for ( let element of students ) {

    // display the values
    console.log(element);
}
```

Comparison

for...of

- The for...of loop is used to iterate through the values of an iterable.
- The for...of loop cannot be used to iterate over an object.

for...in

- The for...in loop is used to iterate through the keys of an object.
- The for...in to iterate over an iterable such arrays and strings but you should avoid using The for...in for iterables.

The While loop

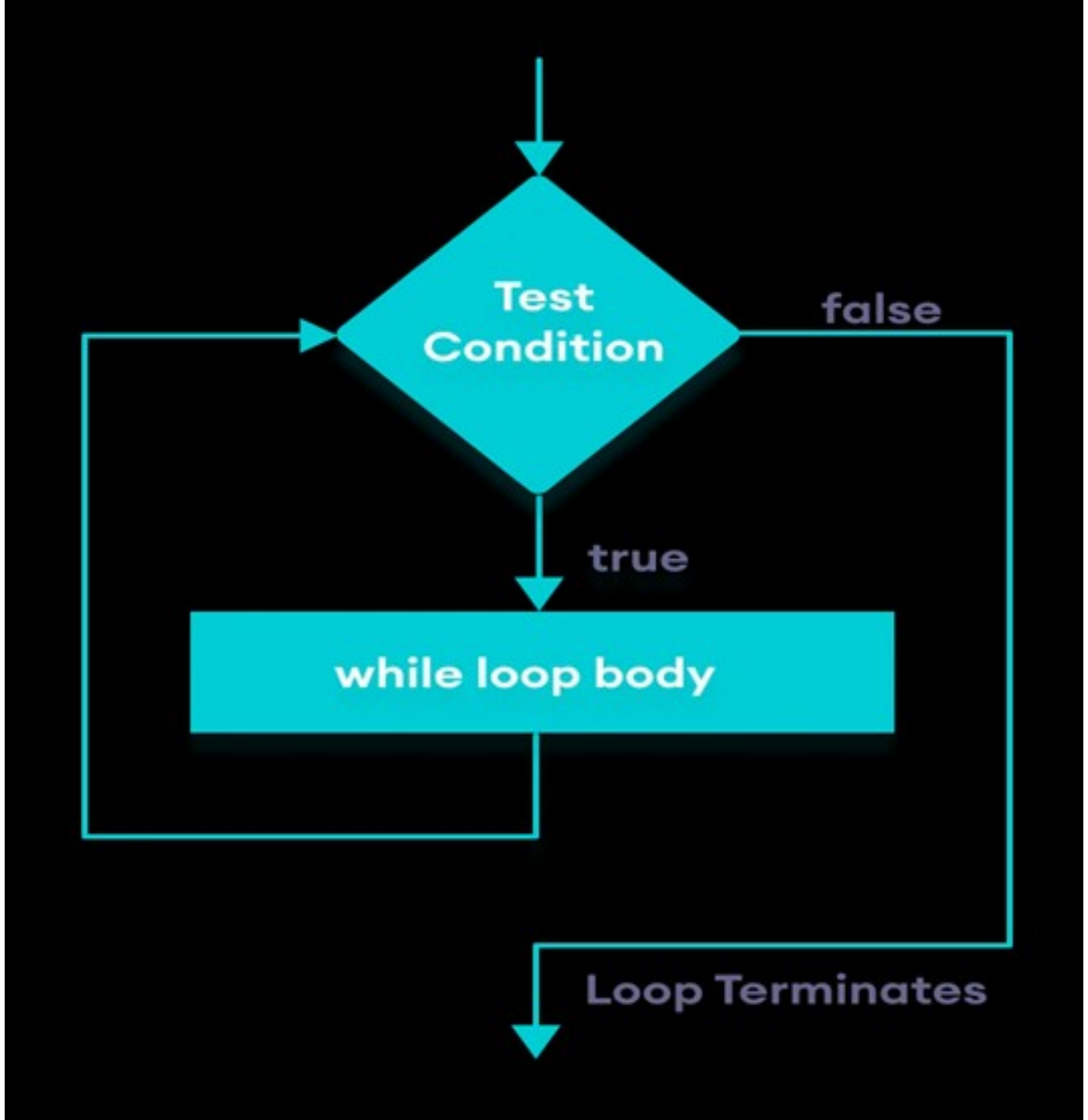
- The syntax of the while loop is:

```
while (condition) {

        // body of loop

}
```

- A while loop evaluates the condition inside the parenthesis ().
- If the condition evaluates to true, the code inside the while loop is executed.
- The condition is evaluated again.
- This process continues until the condition is false.
- When the condition evaluates to false, the loop stops.


Test
Condition
false
true
while loop body
Loop Terminates

Tasks

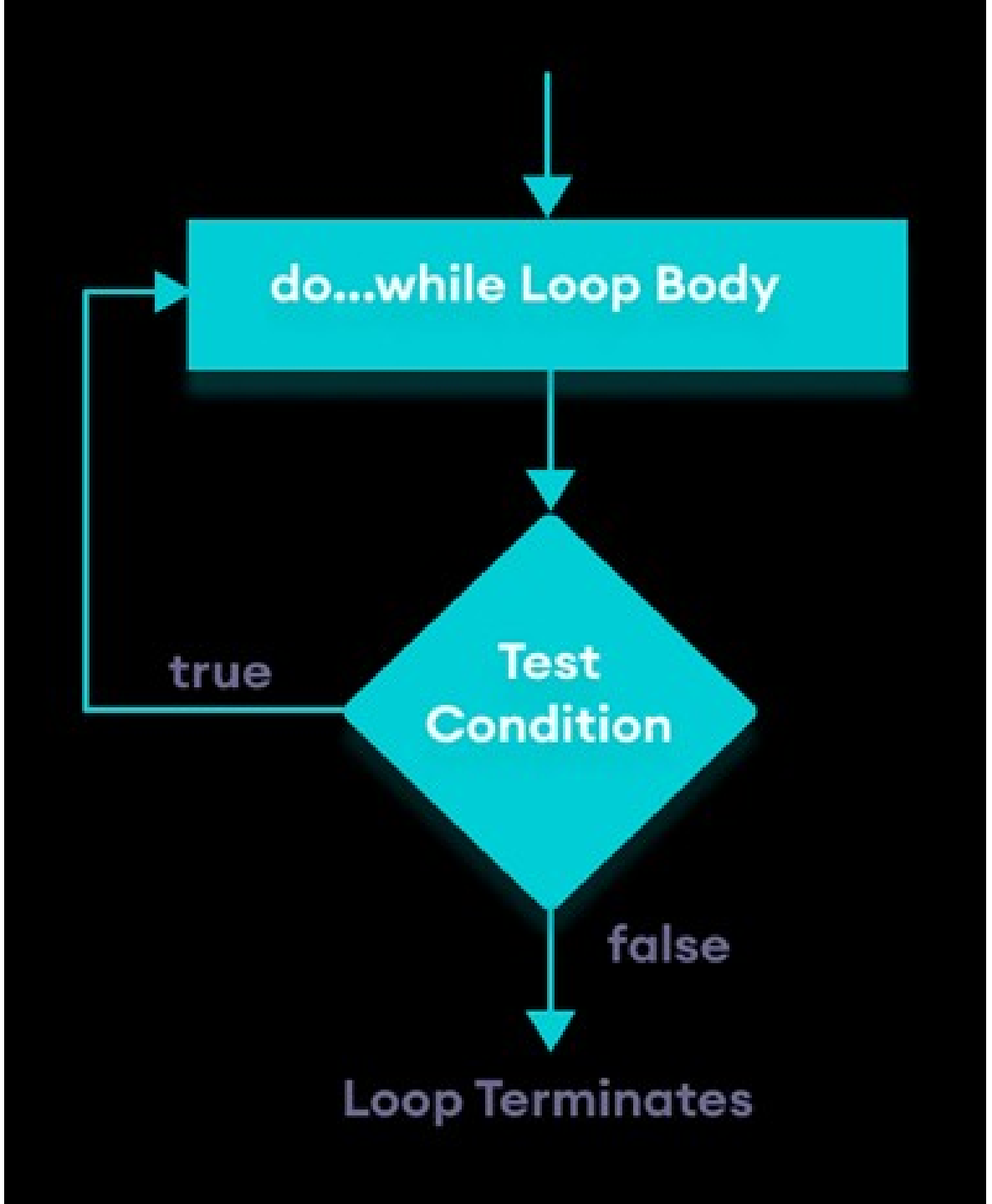
- Example 1: Display Numbers from 1 to 5
- Example 2: Uses the while statement to output the odd numbers between 1 and 10.
- Example 3:Find factorial of a number.

The do-while loop

- The syntax of do...while loop is:

```
do {
    // body of loop
} while(condition)
```

- The body of the loop is executed at first. Then the condition is evaluated.
- If the condition evaluates to true, the body of the loop inside the do statement is executed again.
- The condition is evaluated once again.
- If the condition evaluates to true, the body of the loop inside the do statement is executed again.
- This process continues until the condition evaluates to false. Then the loop stops.

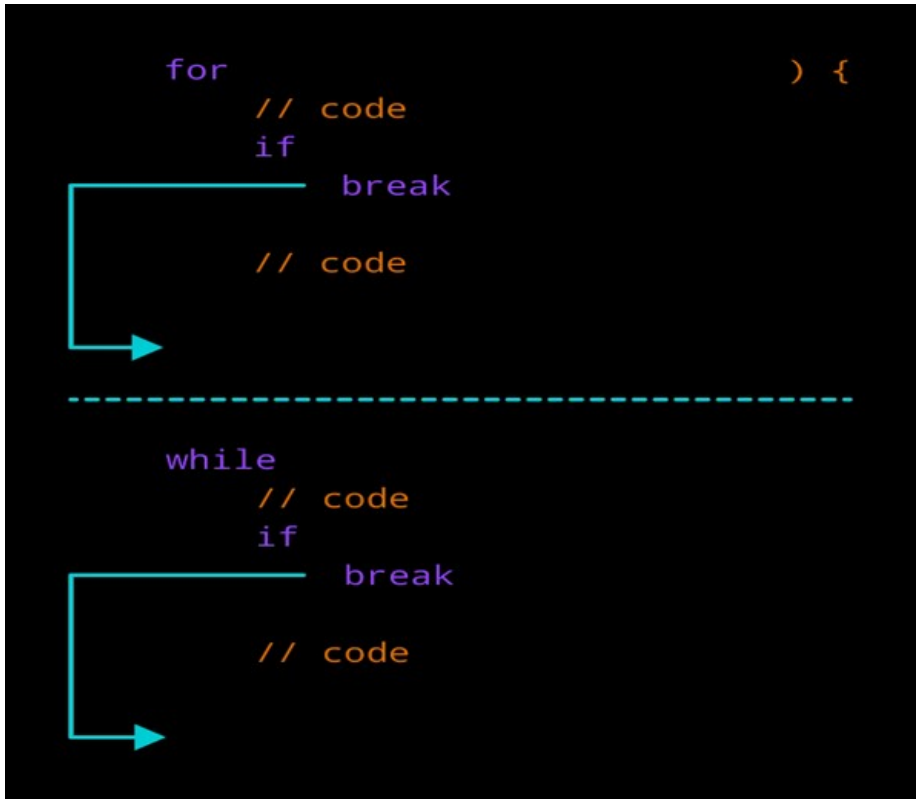

do...while Loop Body
Test Condition
true
false
Loop Terminates

Tasks

- Example 1: Display Numbers from 1 to 5
- Example 2:Multiplication table for 5

break Statement

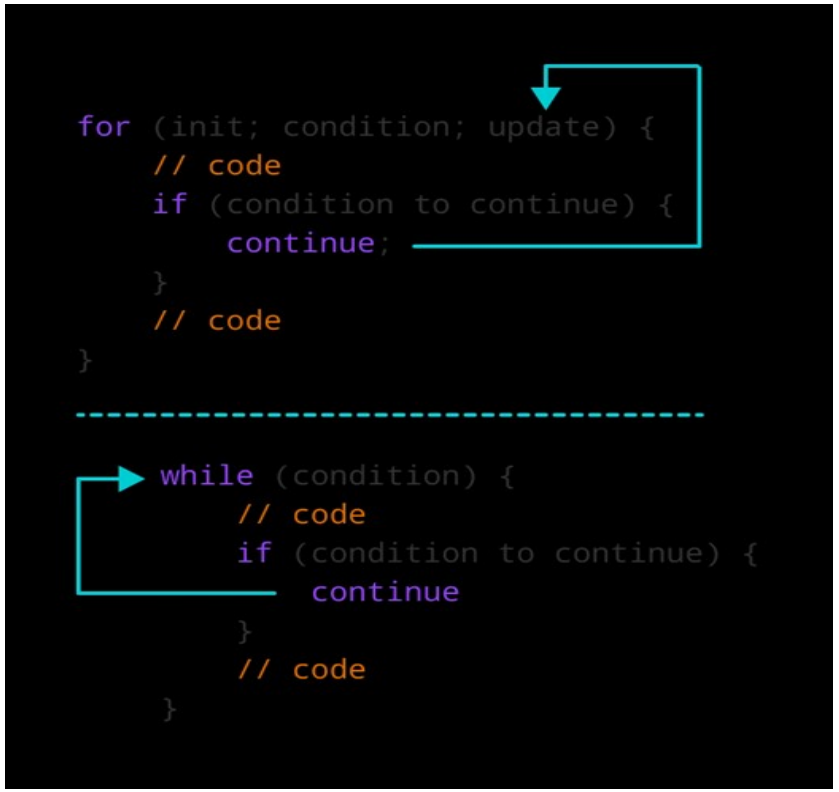
- The break statement is used to terminate the loop immediately when it is encountered.
- The syntax of the break statement is:
- break ;



continue Statement

- The continue statement is used to skip the current iteration of the loop and the control flow of the program goes to the next iteration.
- The syntax of the continue statement is:

  continue ;



Tasks

- Task 1: write a program to print largest of three numbers
- Task 2: Difference between while loop and do while loop.
- Task 3: Difference between break and continue.
- Task 4:multiplication table for 5 using do while
- Task 5:Write a program to print the marks of a student in an object using for loop

  Stud={ Amal=78,Kiran=98,Rahul=87}

Thank you